

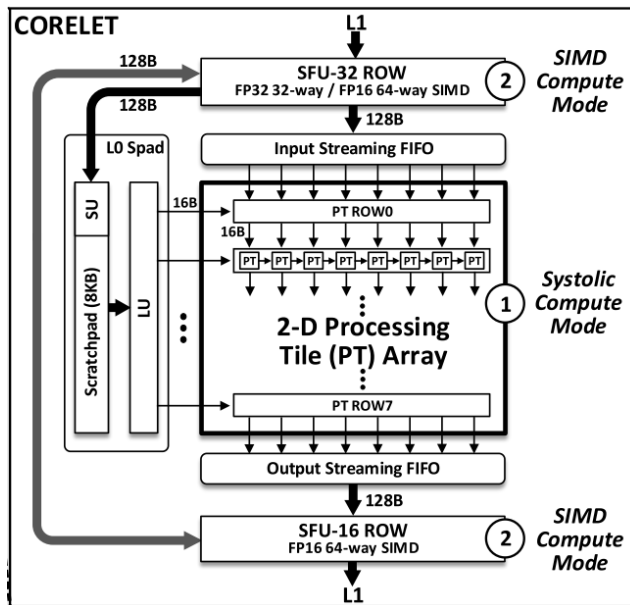
ECE260B Winter

Project Description

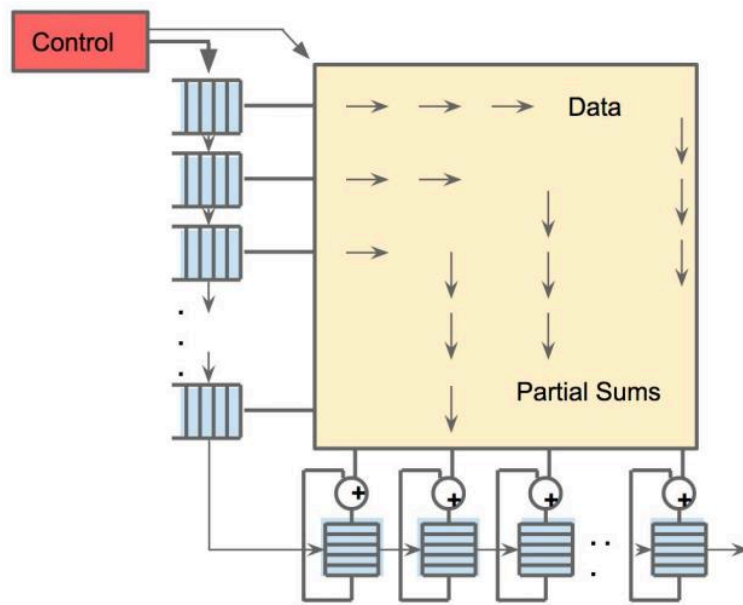
Prof. Mingyu Kang

UCSD Computer Engineering

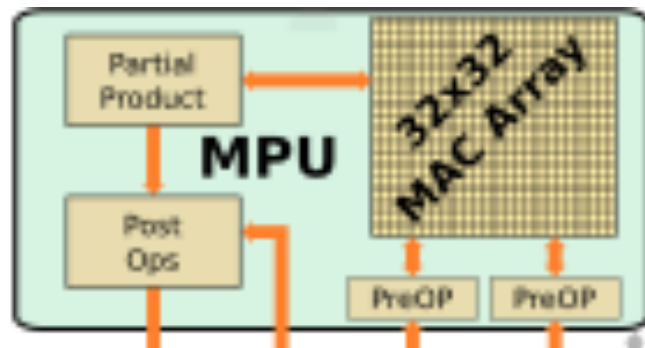
2D Systolic-based Architecture



IBM Rapid



Google TPU



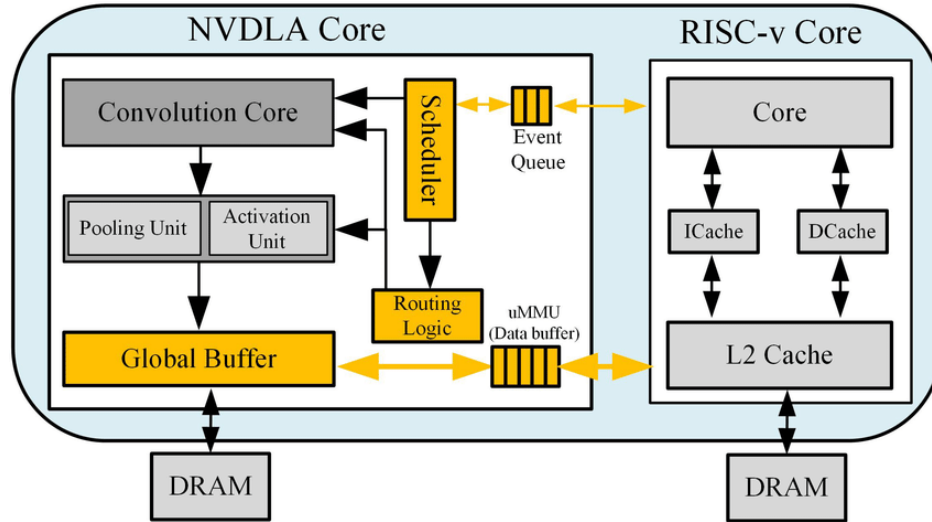
Intel Nervana

J. Oh, "A 3.0 TFLOPS 0.62V Scalable Processor Core for High Compute Utilization AI Training and Inference"

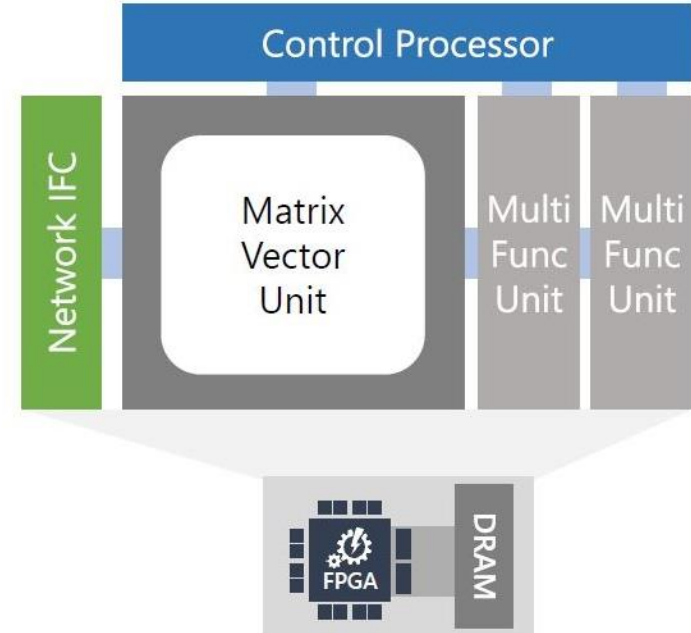
<https://arxiv.org/pdf/1704.04760.pdf>

<https://fuse.wikichip.org/news/3270/intel-axes-nervana-just-two-months-after-launch/>

1-D Vector Processor based Architecture

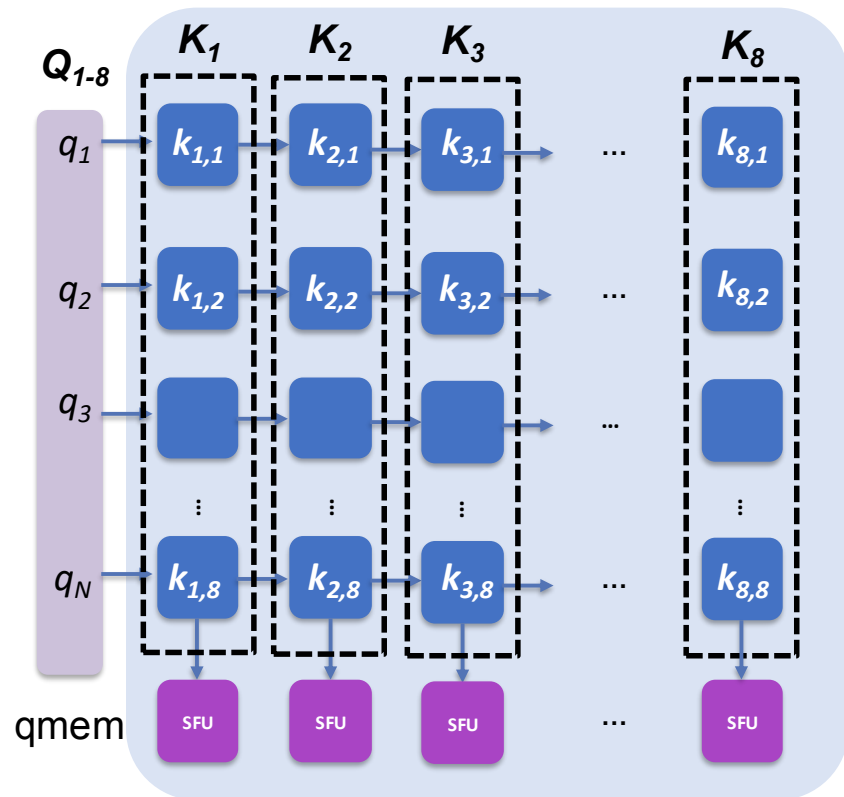


NVIDIA NVDLA

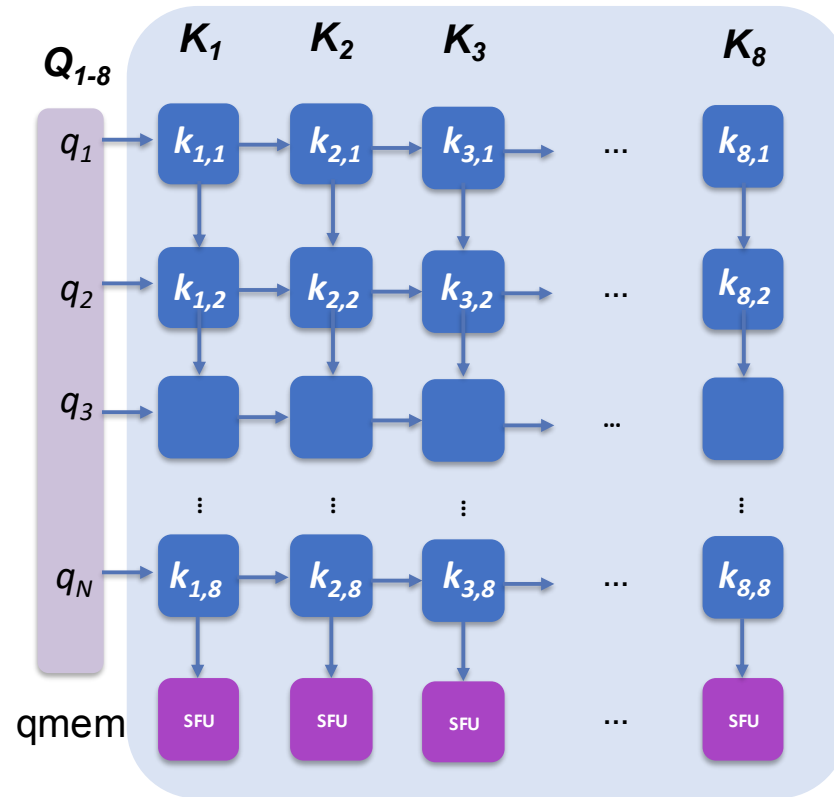


Microsoft Brainwave

Architecture



1-D vector processor architecture



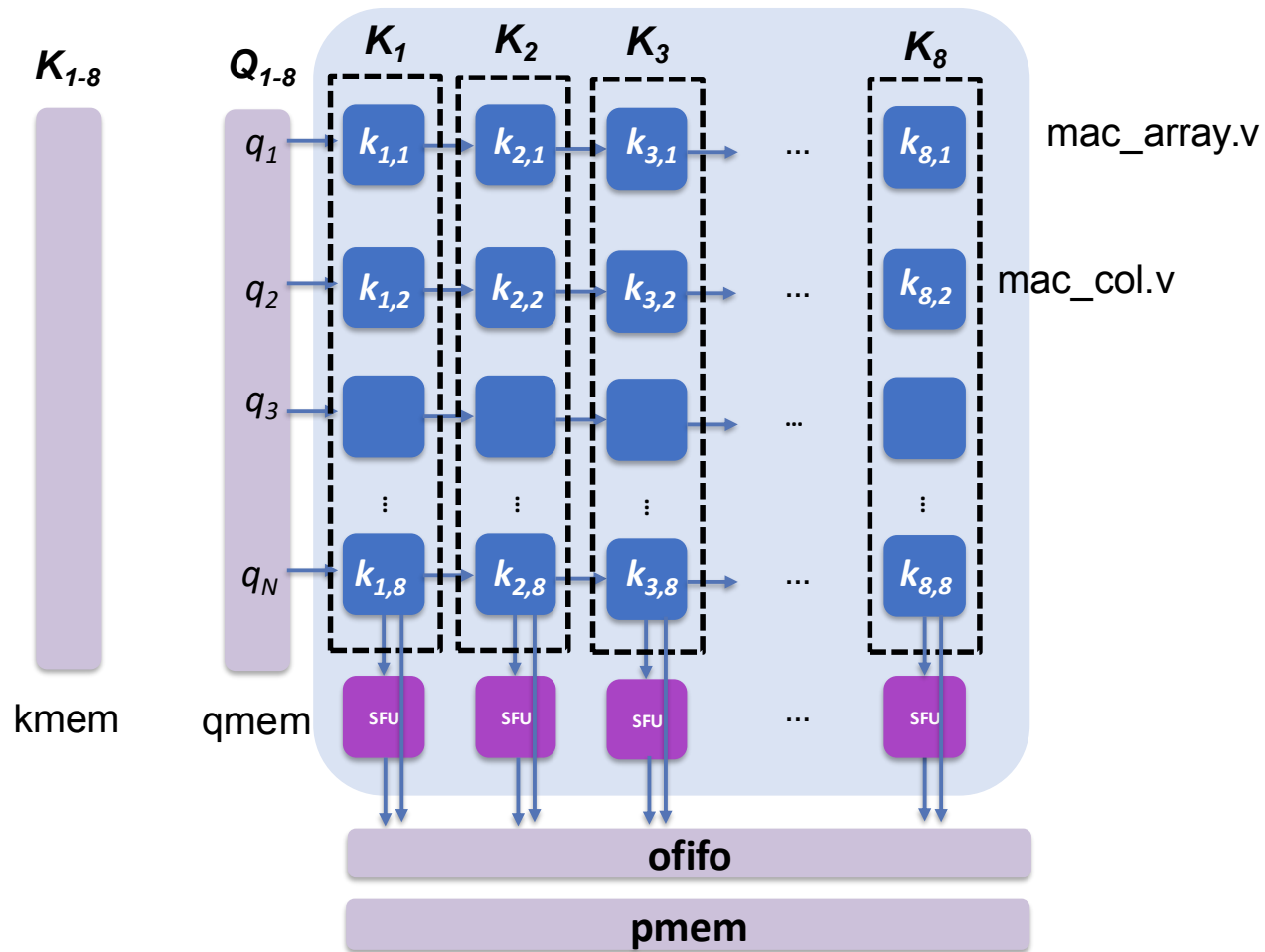
2D-systolic array architecture

Attention Mechanism in Transformer Models

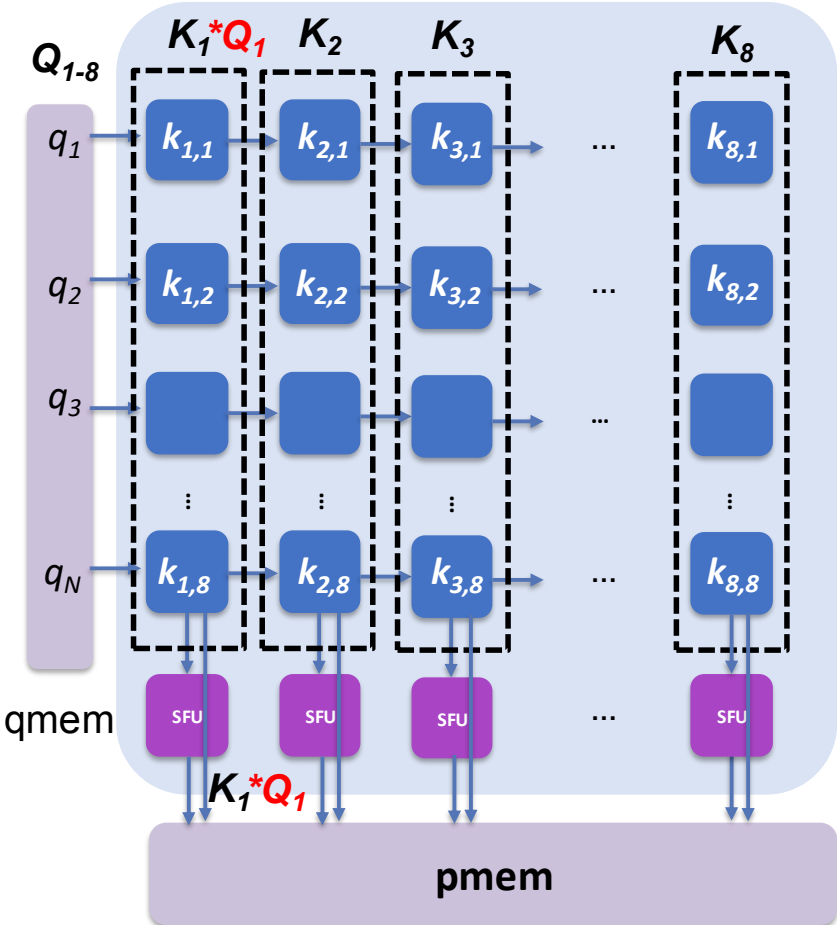
	Morril's	wife	, Ellie,	hugged	her	Sister.
Query (Q)	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6
Key (K)	K_1	K_2	K_3	K_4	K_5	K_6
Value (V)	V_1	V_2	V_3	V_4	V_5	V_6
Score ($= Q \times K$)	0.1	0.3	0.35	0.2	0.15	0.1
Prob ($= \text{Softmax}[QK/\sqrt{d}]$)	0.01	0.41	0.45	0.02	0.1	0.01
Prob * Value (V)	$0.01V_1$	$0.41V_2$	$0.45V_3$	$0.02V_4$	$0.1V_5$	$0.01V_6$
Sum			Z_3			

- Q, K, V are vectors (not scalar)
- Note softmax results are unsigned
- For simplicity, we use normalization instead of softmax
- Generally $Q \times K$ requires lower bit precision while $\text{Prob} \times V$ requires higher bit precision

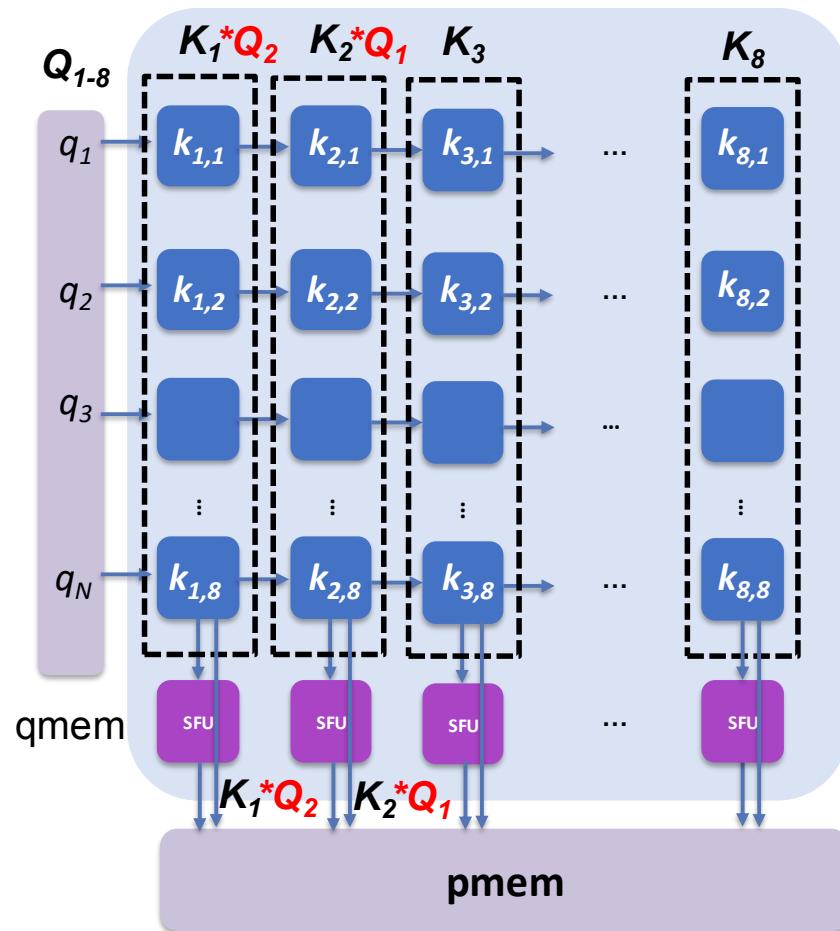
Architecture



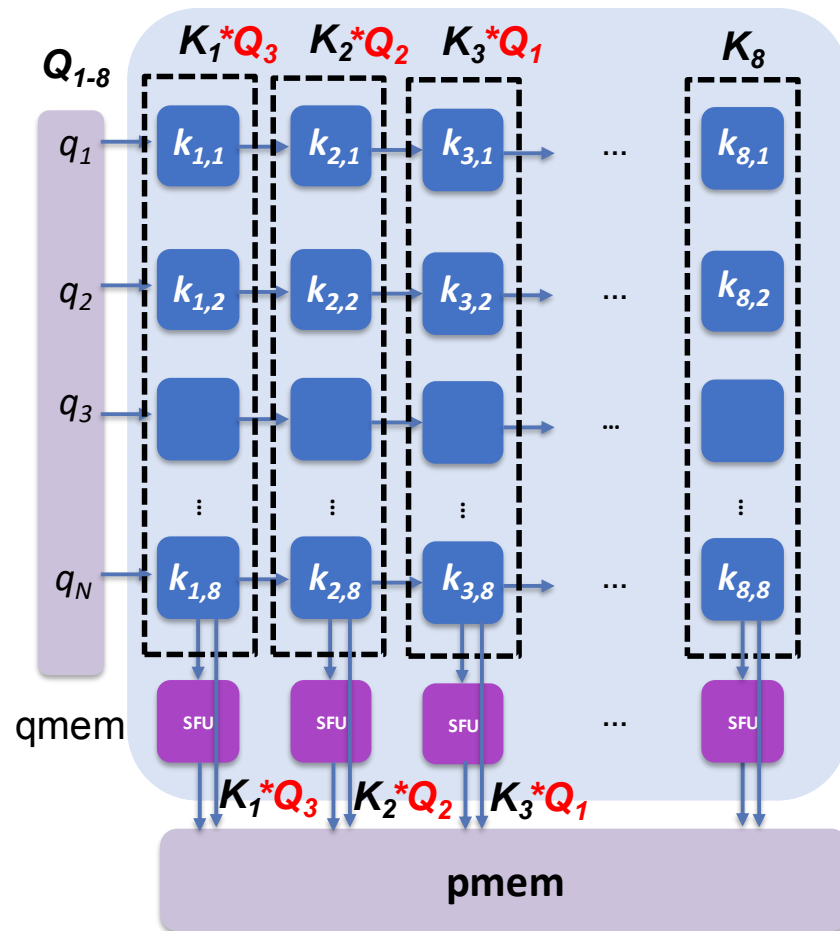
Computation



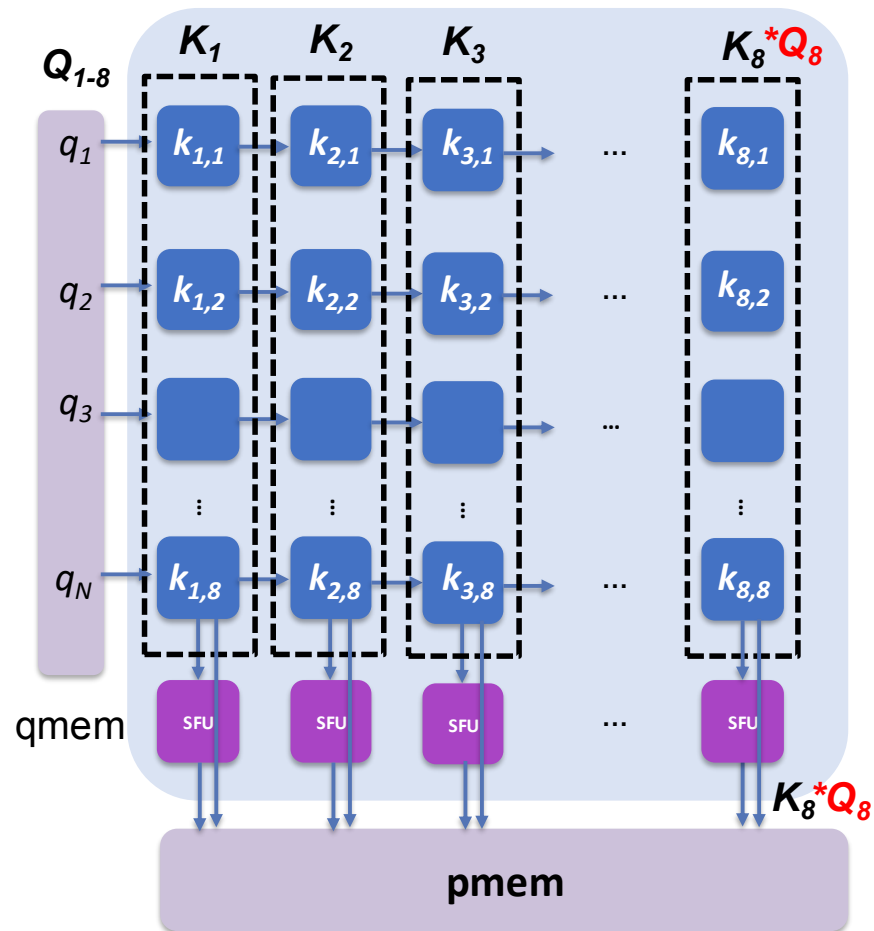
Computation



Computation



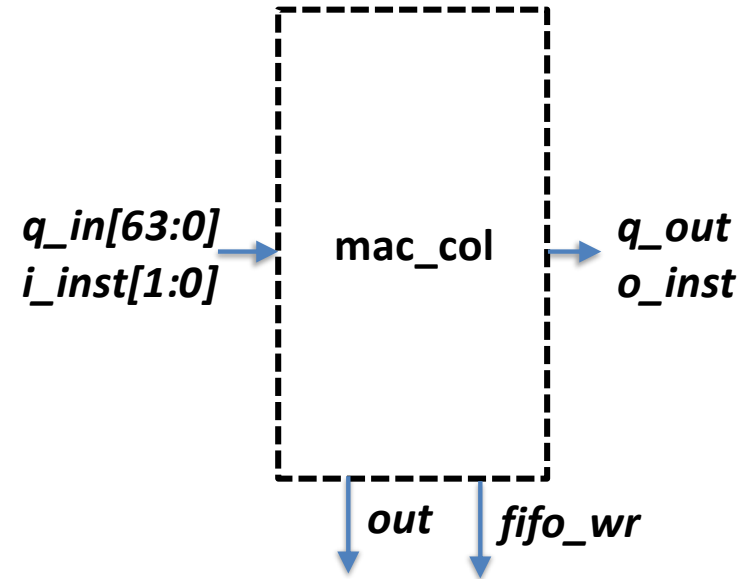
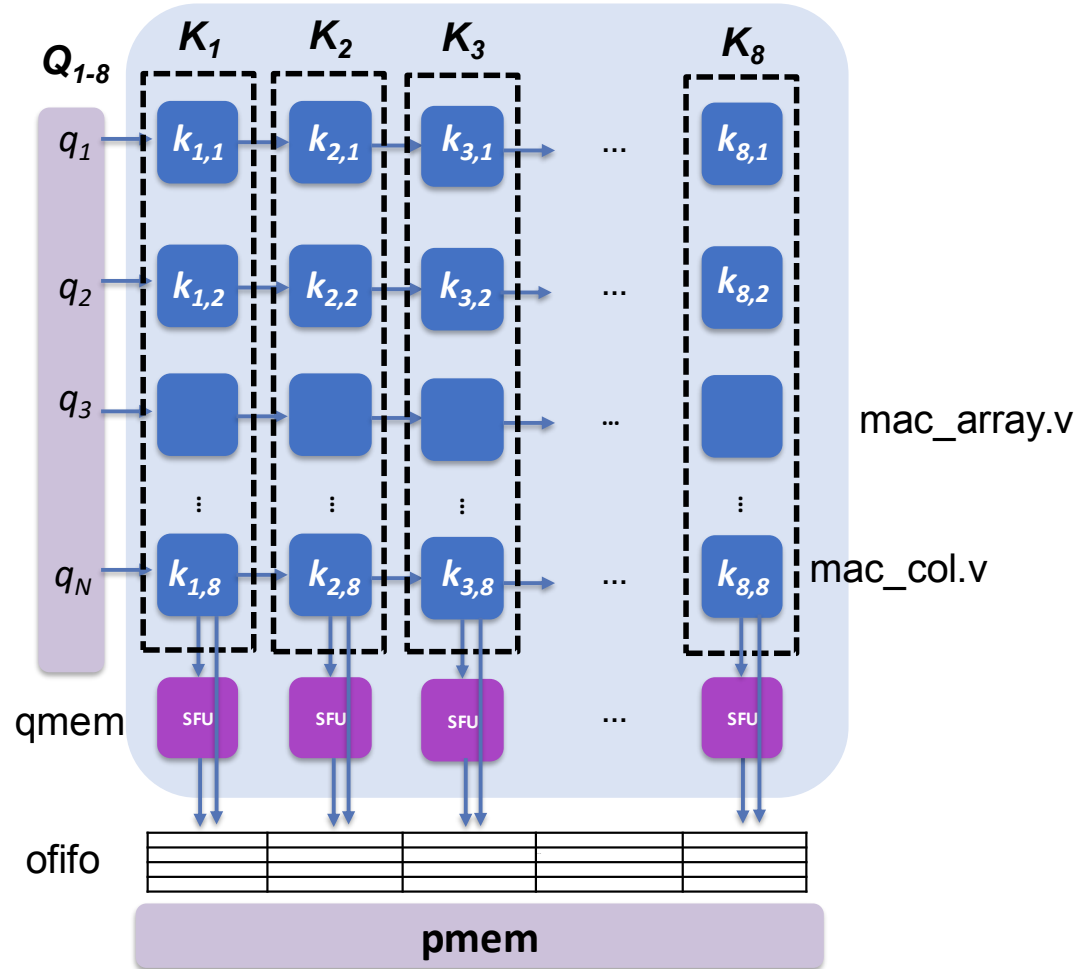
Computation



Processing Stage

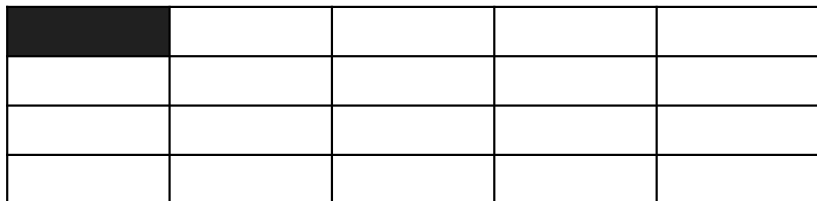
- Stage1: memory write stage
 - qmem & kmem: fill both memory with “qdata.txt” and “kdata.txt”
- Stage2: k-data loading stage
 - move the data from kmem to process’s local register
- Stage3: execute
 - move the data from qmem to input of the processor

More Detail



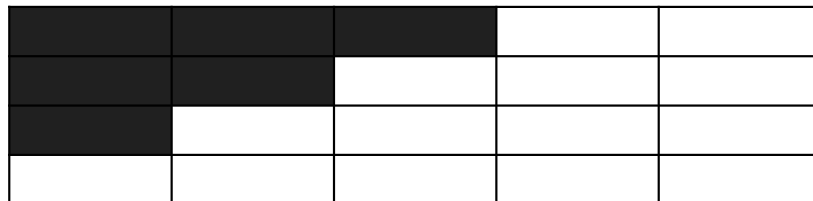
OFIFO Operation

WR = 1



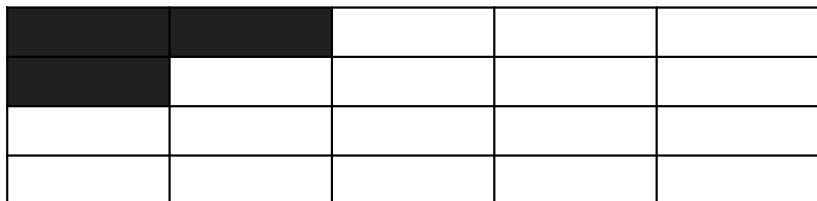
cyc = 1

WR = 1 WR = 1 WR = 1



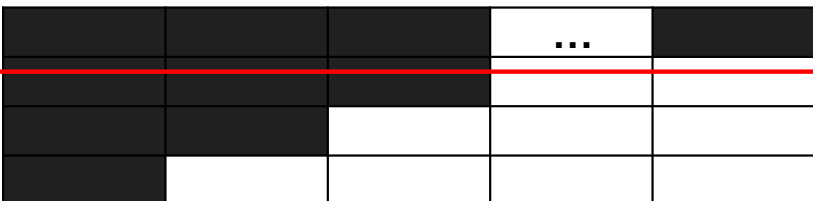
cyc = 3

WR = 1 WR = 1

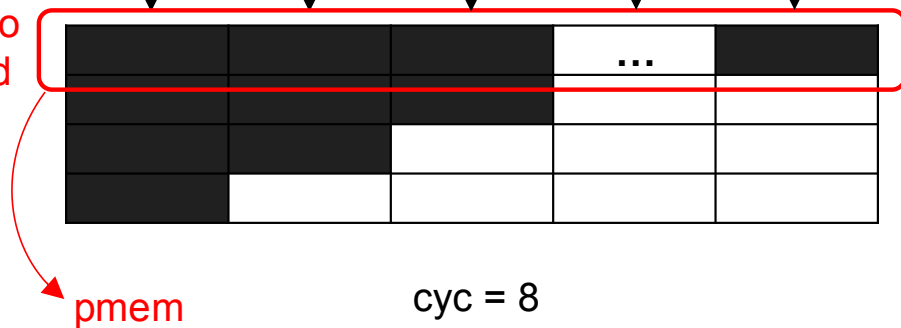


cyc = 2

WR = 1 WR = 1 WR = 1 WR = 1 WR = 1



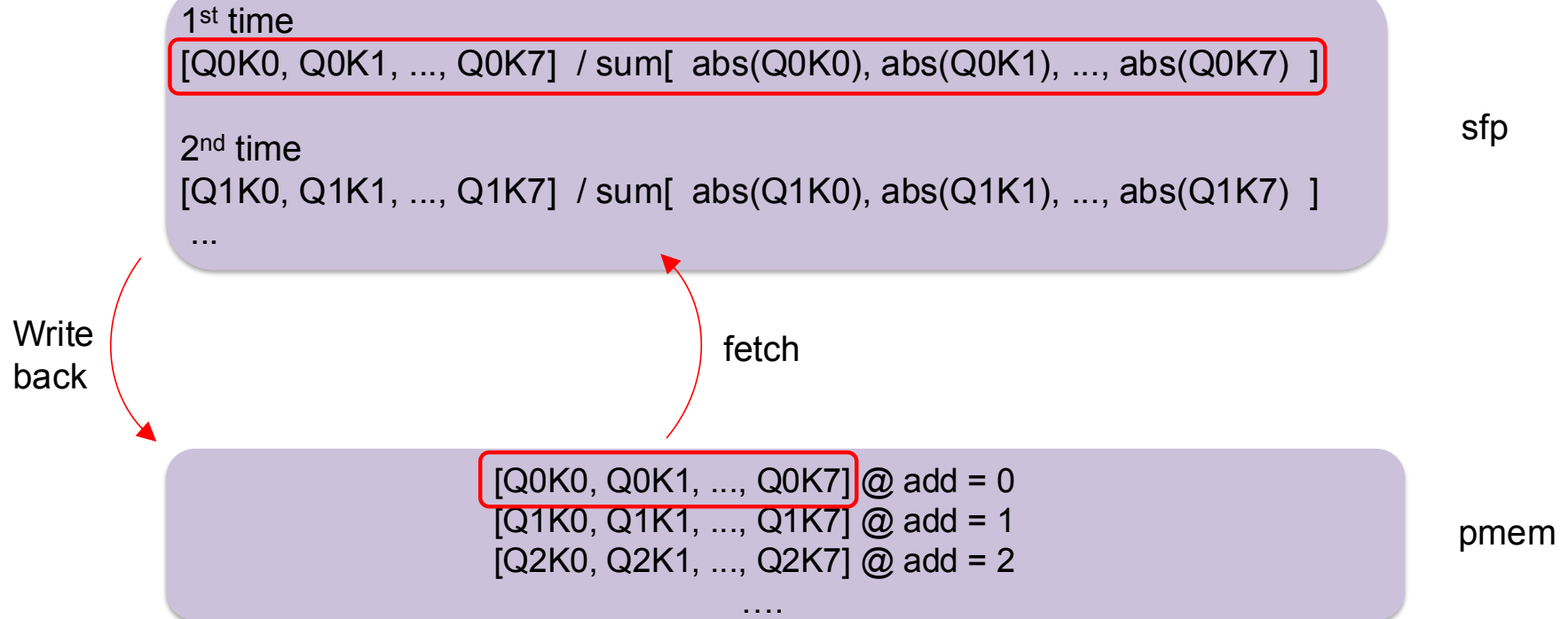
ready to
be read



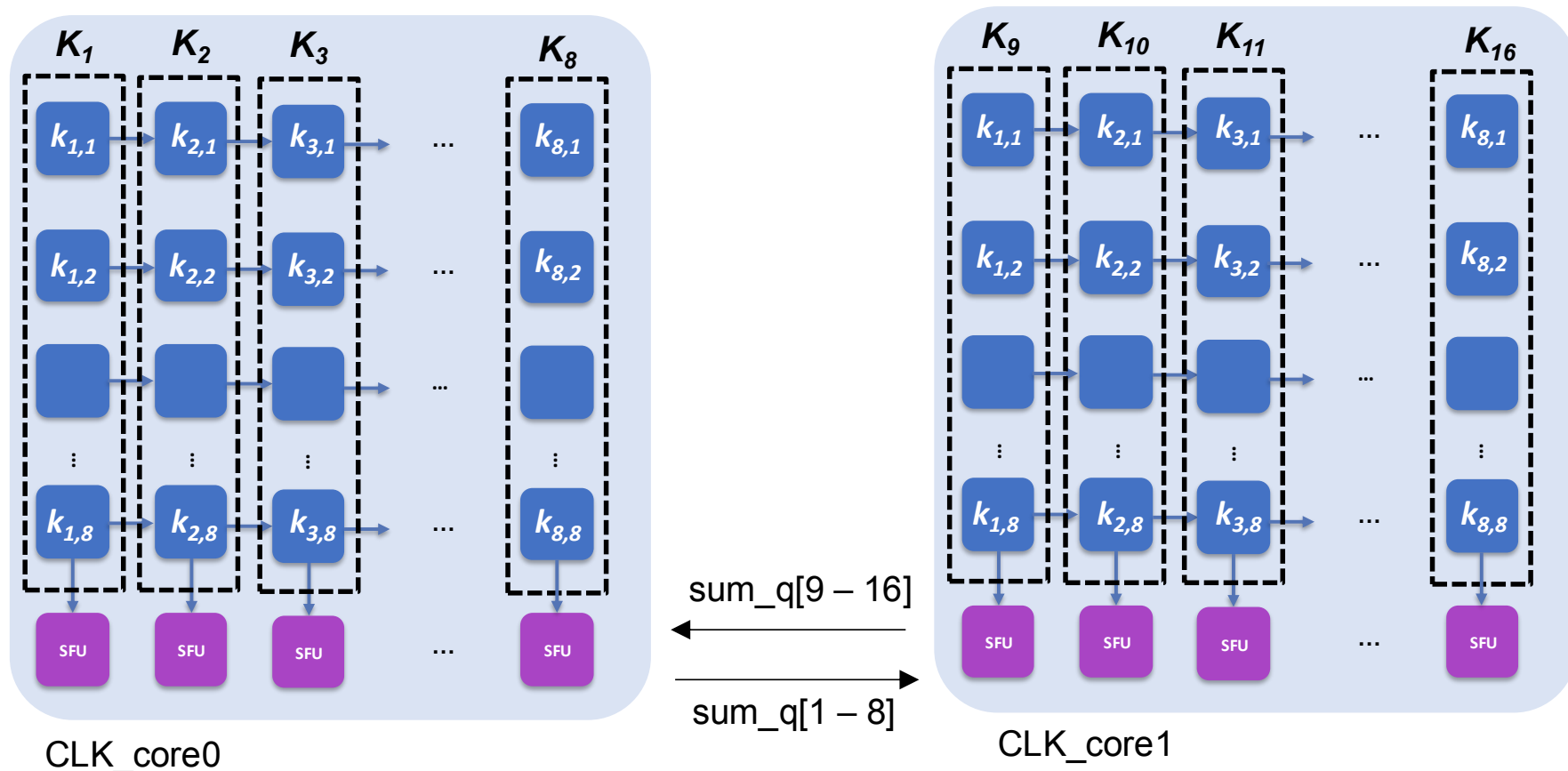
pmem

cyc = 8

Normalization



Extension for Dual Core



Normalize by “ $\text{sum}(\text{abs}(Q_0K_0), \text{abs}(Q_0K_1), \dots, \text{abs}(Q_0K_7))$ from core0
 $+ \text{sum}(\text{abs}(Q_0K_8), \text{abs}(Q_0K_9), \dots, \text{abs}(Q_0K_{15}))$ from core1”

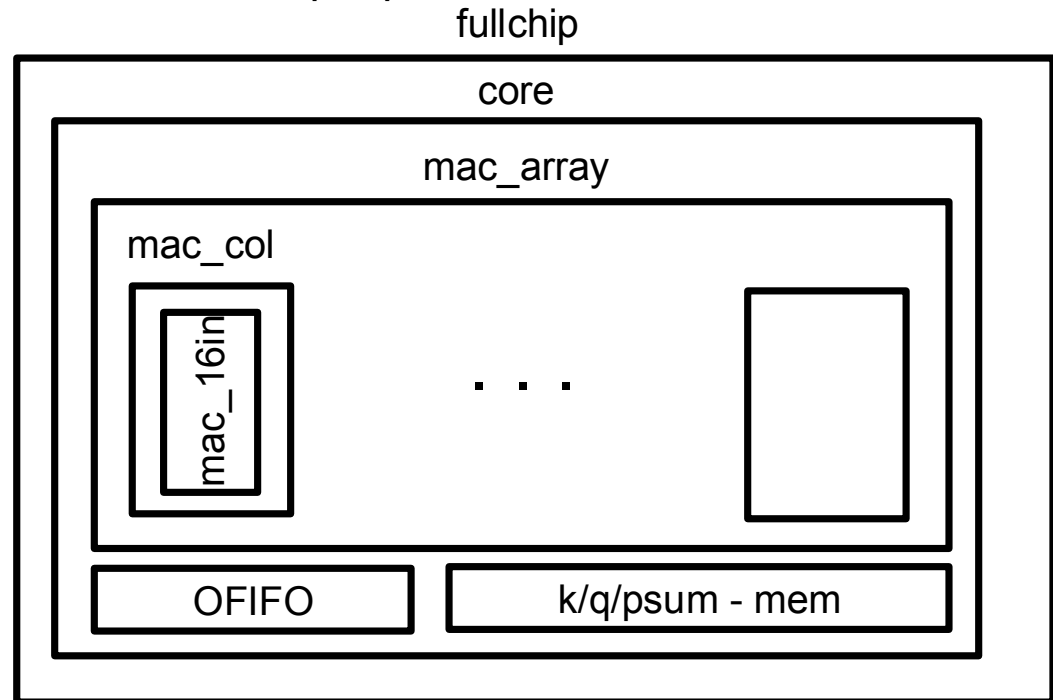
Optimization

- Optimize the implementation to maximize the frequency and minimize power
 - However, the project is not a spec competition
- Apply techniques you have learned from this course
- Potential optimization (does not mean you need to do everything below)
 - Pipeline, multi-cycle path, memory double-buffering, clock gating, loop unrolling, more # of cores, any other creative idea
 - Async interface handling (synchronizer, hand shaking,)
 - Maximize parallelism between processing stages
 - Design your own controller
 - better verification (with random input)
 - Minimize WNS and DRC errors

Understanding Verilog

- Current Verilog includes verification module inside
- Result (pmem_out) of core.v hasn't been connected to output port
- Instructions (fullchip_tb.v)

```
assign inst[16] = ofifo_rd;  
assign inst[15:12] = qkmem_add;  
assign inst[11:8] = pmem_add;  
assign inst[7] = execute;  
assign inst[6] = load;  
assign inst[5] = qmem_rd;  
assign inst[4] = qmem_wr;  
assign inst[3] = kmem_rd;  
assign inst[2] = kmem_wr;  
assign inst[1] = pmem_rd;  
assign inst[0] = pmem_wr;
```



How to prove ?

- Prove the efficacy of your idea (why ? how ? how much ?)
 - measure power and check functionality with VCD
(SDF is not required, but it will be plus alpha)
 - clear comparison before vs. after
 - clear description in the report regarding your design choice
e.g., why you chose the decision, why you cut the pipeline boundary,
 - report is VERY important

Project Policy

- TA & Instructor **will not**
 - solve your project instead of you
 - debug your code (e.g., I got this error with screenshot, can you check my code ?)
 - analyze your issue (e.g., my performance is not so good, do you know why ?)
 - comment on your result (e.g., Can you check my result is correct ?, Can I get A grade with this optimization ?)
 - “fighting against errors” is the biggest part of this project
- However, TA & Instructor **will**
 - clarify the project problem
 - help any environment issue (e.g., it seems there is no license file, tool does not load, ...)
 - provide general guideline for a certain issue
- You can still discuss between students through Piazza